PORTFOLIO · FABRICATION & MECHANICAL

JOHN NORTHRUP

Designer & Digital Fabrication Specialist | Computational Workflows & Physical Systems

I am a designer and technical fabricator specializing in computational modeling, digital workflows, and physical construction. Grounded in hands-on material fabrication—from fine joinery to structural welding—and technical spatial design, my work bridges complex digital systems with buildable, physical space. I focus on technical design development, project management, and translating complex digital models into precise physical outcomes.







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FABRICATION & MECHANICAL

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01

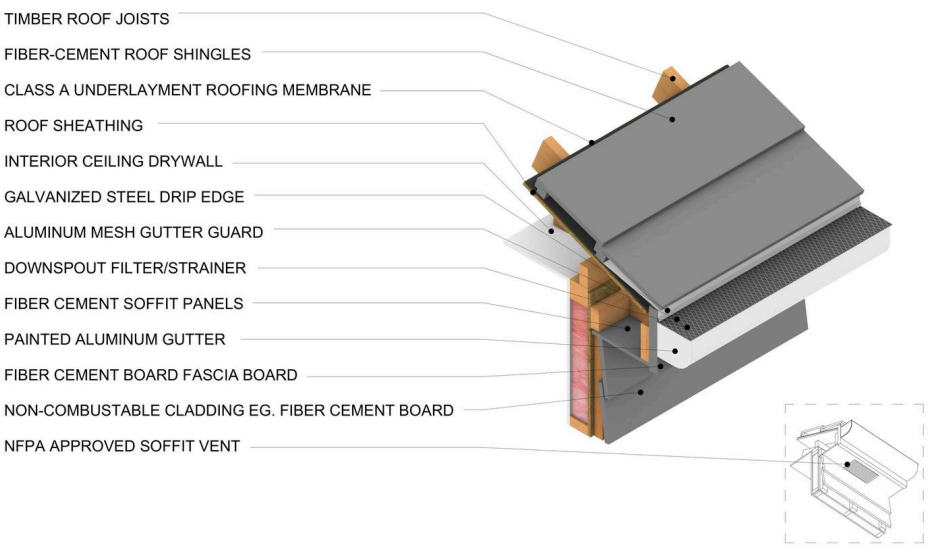
RESILIENT FUTURES

2025, SCI-ARC RESILIENT FUTURES TASK FORCE
ROLE · DRAWINGS, DESIGN & FABRICATION

SCI-Arc formed the Resilient Futures Task Force after the January 2025 firestorms that burned through Altadena and the Pacific Palisades. One thing the fires left behind was suspicion of timber: people rebuilding wanted to know whether building in wood had made it worse.

These drawings and models were made for the task force's public work on that question, alongside its Demystifying Timber session. Nothing in them is exotic. They are the ordinary parts that make an ordinary timber house more likely to survive a fire, and the whole point is that you can see every one. Three conditions are drawn as annotated axonometrics and then built at full scale: the eaves and overhang, the wall, and the deck. I drew all three, and helped design and build the eaves and overhang.

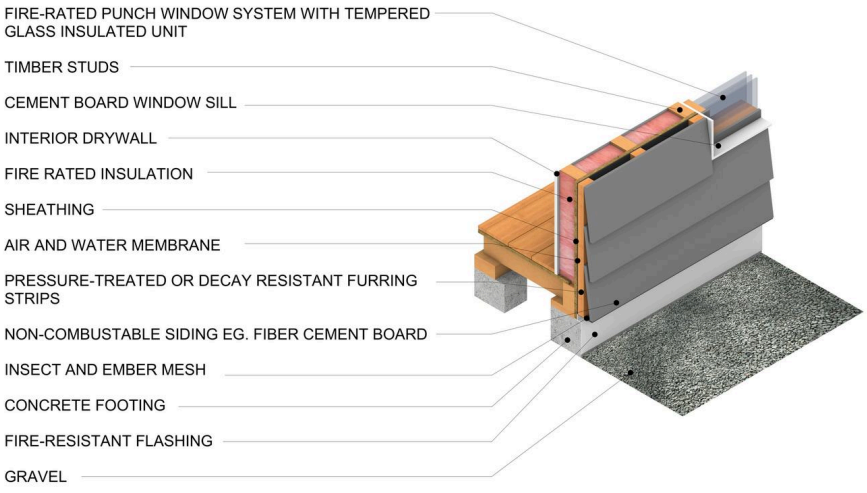
The specification is the argument. Fiber-cement shingles over a Class A underlayment membrane, non-combustible fiber-cement cladding, soffit and fascia, NFPA-approved soffit vents, an aluminum mesh gutter guard and a downspout strainer to keep embers out of the gutter, fire-rated insulation, and a fire-rated window with tempered insulated glass. Built at full size, the vent screen and the gutter guard stop being line weights and become things you can stand in front of.



DRAWING | EAVES AND OVERHANG, LAYER BY LAYER



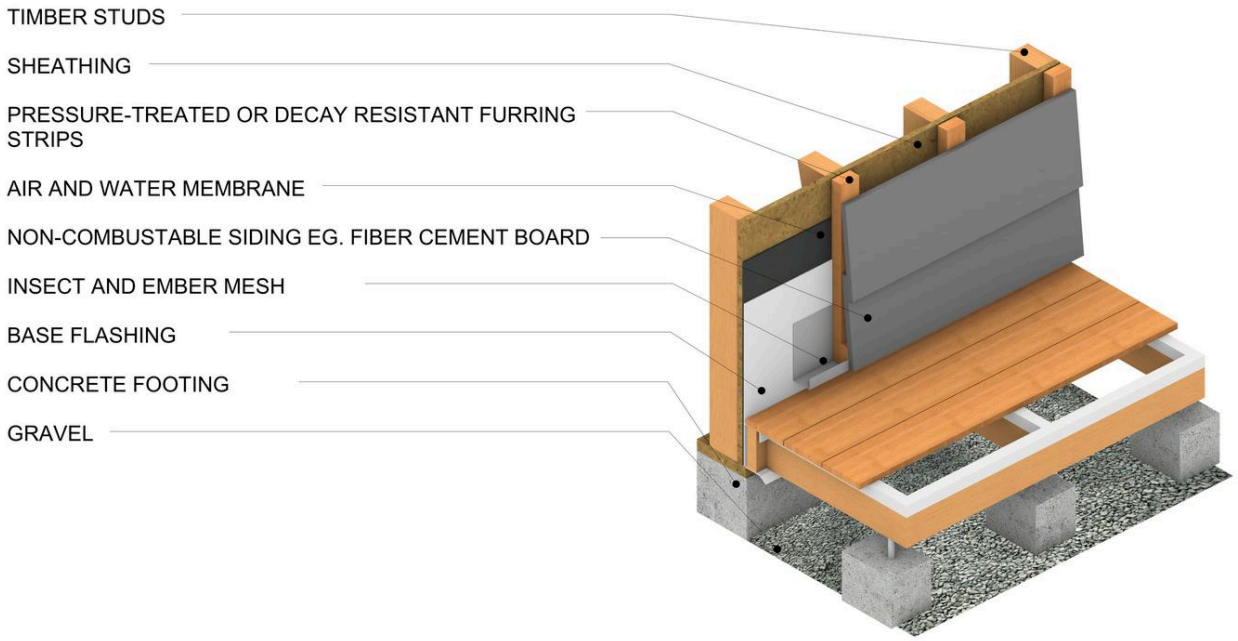
MODEL | THE VENTED SOFFIT AND GUARDED GUTTER, FROM BELOW



DRAWING | WALL, WITH THE FIRE-RATED WINDOW AND SILL



MODEL | THE TIMBER FRAME BEHIND THE CLADDING



DRAWING | DECK, FURRING AND NON-COMBUSTIBLE SIDING



MODEL | THE ASSEMBLY AT FULL SCALE



02

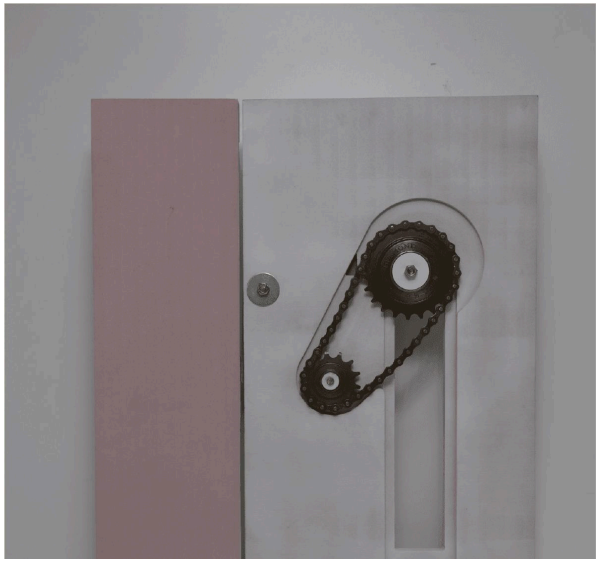
DETAILS, DETAILS

F25, DETAILS, DETAILS, WITH CHIA-CHEN LI AND BAIZHEN YU
ADVISOR · DWAYNE OYLER

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.



FOAM PROTOTYPING PROCESS



PROTOTYPE INTERIOR MECHANISM



FINAL DOOR MECHANISM, CLOSE-UP



03

ELF-O-MATIC-3000

FA22, MONSTER CITY STUDIOS
ROLE · TECHNICAL DESIGNER & PROJECT MANAGER

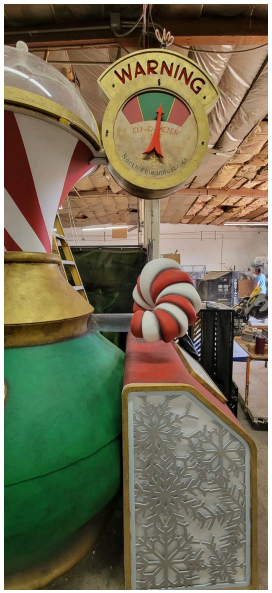
A holiday show piece built at Monster City Studios: a steam-powered candy machine with a neon marquee, a lit viewing window of tumbling balls, a STEAM grille, a pressure gauge and a candy-striped dome. Behind the theming is a working show, with lighting, moving parts and an operator panel to reset it between guests.



BUILT WORK | THE MACHINE RUNNING IN THE SHOP (VIDEO: JCNORTHRUP.COM/PROJECTS/E...



BUILT WORK | MARQUEE, GAUGES AND DOME



DETAIL | WARNING GAUGE



DETAIL | OPERATOR PANEL



DETAIL | SERVICE HATCH WITH LABELED CONTROL SWITCHES